

TASK 2 — RESEARCH ON CLOTHING FITTING ALGORITHMS

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1. Purpose of the Research

The purpose of this research was to understand how Artificial Intelligence can digitally analyse human body structure and automatically adjust virtual clothing on users.

This study focused on:

- smart garment fitting systems
- digital body analysis
- AI-assisted clothing alignment
- virtual dressing technologies
- intelligent fashion applications

The task also explored how computer vision improves online shopping and digital fashion platforms.

2. Overview of Intelligent Garment Fitting Systems

Intelligent garment fitting systems are AI-powered applications that simulate clothing on digital human models.

These systems use:

- body landmark analysis
- body contour estimation
- motion tracking
- AI clothing adjustment
- computer vision processing

The technology helps customers preview clothing digitally before purchasing products online.

Industries using these systems include:

- fashion retail
- e-commerce platforms
- virtual fashion design
- augmented reality shopping
- AI styling applications



Figure 1 — AI Clothing Fitting Technology

3. Functional Workflow of Digital Clothing Fitting

The workflow of AI garment fitting systems consists of several processing stages.

Stage 1 — Person Identification

The computer vision system first identifies the human body from the image or video frame.

The AI system detects:

- upper body
- lower body

- arms
- legs
- body posture

Stage 2 — Pose Landmark Mapping

Pose estimation frameworks generate body landmark points.

The AI tracks:

- shoulder positions
- waist alignment
- hip positioning
- limb orientation

This helps the system understand body proportions.

Stage 3 — Digital Body Dimension Analysis

The system estimates body dimensions digitally using:

- landmark distance calculation
- contour estimation
- body scaling techniques

Estimated measurements include:

- shoulder width
- waist area
- body height
- torso dimensions

Stage 4 — Garment Positioning

Virtual clothing is aligned on the body according to:

- body posture
- body shape
- movement direction
- body orientation

Stage 5 — Dynamic Visualization

The final clothing simulation is displayed dynamically with body movement support.

```
Fetching image from URL: https://cdn.dsmcdn.com/mnresi
Image downloaded. Size: 61371 bytes
Converted image to RGB.
Initializing Pose and FaceMesh models...
Pose and FaceMesh models initialized.
Face fully visible.
Pose landmarks detected. Drawing landmarks...
```



Figure 2 — Human Body Landmark Tracking

4. Technologies Used in Clothing Fitting Systems

Technology	Purpose
Artificial Intelligence	Smart fitting analysis
Computer Vision	Body tracking
Pose Estimation	Landmark detection
OpenCV	Image processing
MediaPipe	Body landmark tracking
Deep Learning	Clothing prediction

5. Practical Applications of Garment Fitting Systems

AI clothing fitting systems are used in:

- virtual fitting rooms
 - online shopping platforms
 - AI fashion assistants
 - digital fashion systems
 - augmented reality applications
 - personalized shopping systems
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6. Benefits of AI Garment Fitting Systems

The advantages include:

- reduced clothing return rates
- improved shopping experience
- personalized recommendations
- realistic clothing visualization
- faster product selection
- better fitting estimation

7. Limitations and Technical Issues

Challenge	Description
Poor Lighting	Reduces body visibility
Loose Clothing	Hides body structure
Complex Background	Reduces tracking accuracy
Fast Movement	Causes instability
Incorrect Measurements	Affects fitting quality

8. AI Models Used in Clothing Fitting

Popular AI frameworks include:

- MediaPipe
- OpenPose

- TensorFlow
- OpenCV
- Deep Learning Networks

These systems help improve:

- pose tracking
- body measurement estimation
- clothing alignment
- real-time fitting

8.1 Algorithm — AI Clothing Fitting Workflow

Step 1

Capture human body image or webcam frame.

Step 2

Detect the human body using computer vision techniques.

Step 3

Identify body landmarks such as:

- shoulders
- elbows
- hips
- knees
- ankles

Step 4

Estimate body dimensions using landmark positions and body proportions.

Step 5

Load the digital clothing model or garment image.

Step 6

Align the clothing according to:

- body posture
- body shape
- movement direction

- body orientation

Step 7

Adjust clothing size and fitting dynamically.

Step 8

Render the final virtual clothing simulation.

Step 9

Display the output in real-time.

Step 10

Repeat the process continuously for movement tracking.

8.2 Pseudo Code — AI Clothing Fitting System

START

Capture image or webcam frame

Detect human body

Extract body landmarks

Estimate body dimensions

Load virtual clothing model

Align clothing with body posture

Adjust clothing fitting and scaling

Render clothing on human body

Display final virtual fitting output

END

9. Knowledge Gained

During this task, the following concepts were learned:

- AI clothing fitting workflows
- pose landmark tracking
- digital fitting systems
- body measurement estimation
- AI fashion technology

- computer vision applications
 - virtual fitting systems
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10. Final Summary

AI clothing fitting algorithms are transforming modern fashion technology using Artificial Intelligence and computer vision systems.

The research improved understanding of:

- AI fitting systems
- body landmark tracking
- digital clothing simulation
- virtual fitting rooms
- AI shopping systems
- pose estimation workflows

These systems play an important role in improving online shopping experiences and intelligent fashion recommendation platforms.